



Research Article

Extended State Observer based control for coaxial-rotor UAV



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ABSTRACT

This paper considers the problem of controlling the position and the orientation of a Coaxial-Rotor Unmanned Aerial Vehicle -CRUAV- despite unknown aerodynamic efforts. A hierarchical flight controller is designed, allowing the trajectory tracking and the stabilization of the vehicle. The designed controller is built through a hierarchical approach yielding two control loops, an inner one to control the attitude and an outer one to control the translational trajectory of the rotorcraft. An Extended State Observer -ESO- is used to estimate the state and the unknown aerodynamic disturbances. The analysis further extends to the design of a control law that takes the disturbance estimation procedure into account. Numerical simulations are carried out to demonstrate the efficiency of the proposed control strategy.

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1. Introduction

Recently, an increasing interest in developing autonomous unmanned aerial vehicles (UAVs) as exemplified by small-scale (HUAV) [13,14] and multirotor micro-aerial vehicles (MAV) [1–18] capable of achieving a range of applications in both civil and military scenarios that can be addressed with such aerial systems like surveillance, reconnaissance, aggressive maneuvers in complex and dangerous environments. Current research goals include the development of specialized flying machines equipped with autonomous control devices capable of performing quasi-stationary as well as forward flight with high speeds and accelerations to reach remote areas. Miniature rotorcraft (RUAVs) are motivated by crucial characteristics such as small size, autonomous navigation capacity, capability of vertical takeoff and landing (VTOL), flight at very low altitudes and high maneuverability. Several different aerodynamical configurations are available according to their mechanical configuration, some have two rotors like the coaxial helicopter [3–6], coaxial rocket helicopter [9], twin rotor system [15], ducted fan VTOL [17], others have three rotors like the trirotor [16] and some others have four rotors, they are called quadrotor MAV [2,8,11].

The coaxial-rotor helicopter is a small rotorcraft having two main rotors and driven by two brushless DC motors which rotate in opposite direction in order to overcome gyroscopic torque and cancel the drag torque produced by each rotor in case of hover or quasi-

stationary flight [5]. The yaw torque is obtained by a differential speed variation between the two rotors, whereas the altitude is controlled by simultaneously varying the rotor speeds. In order to ensure the flight of the vehicle in all directions in the space, a cyclic swash plate is placed on the lower rotor, and is actuated by two servo motors in order to produce the inclination of the lower rotor to generate roll and pitch torques due to disc rotor tilting of the lower rotor. Previous studies [6,9,10,17] have demonstrated the CRUAV concept feasibility based on theoretical and experimental investigations concerning the sizing of the device, the payload and autonomy.

In the literature, various linear and nonlinear controllers have been developed to achieve flight performances of aerodynamic rotorcraft configurations. Some linear controllers have been applied to RUAVs with success based on PD controllers [16], PID controllers [13], and H_∞ Controllers [4]. However, linear controllers fail to provide good flight performance at non-nominal operating conditions such as during high-speed and aggressive flights. More recently, interest has been focused on nonlinear control laws for mini rotorcraft vehicles. One such system, based on the quadrotor airframe, has demonstrated successful flight using adaptive backstepping control in [8]. PID controllers was presented in [12], leading to a time scale separation between the cascaded systems [19], sliding mode disturbance observer designs [18], using dynamic extension of the thrust input, or the control of a quadrotor aircraft via singular perturbations [27]. In the coaxial-rotor project known as Gun Launched MAV [3,6,10], a hierarchical backstepping was applied to design a nonlinear controller for a miniature coaxial vehicle with experimental results [5], approximate feedback linearization was introduced in [14] ignoring the input coupling. In [9],

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a sliding mode control has been applied to design a nonlinear controller for coaxial rocket helicopter and a disturbance observers based control is applied to UAVs in [7,15] with experimental results.

Active Disturbance Rejection Control (ADRC) is a well known technique extensively used for control of nonlinear systems [20,26,32]. For example, it has been applied to helicopter trajectory tracking [14], VTOL vehicles [23], wing rock motion control [21], tactical missiles [22] and hypersonic reentry vehicle [20]. The ADRC proposes a simple (yet very powerful) framework, which can be used for a variety of control problems [28]. The main idea behind it is that the unknown plant dynamics, uncertainties and disturbances are actively rejected for the same in the control law in real-time. This lumped perturbation, also denoted as a total disturbance, is treated as an additional state variable of the process, seen from the system point of view as an extra input signal. The augmented state is estimated online with an observer in an inner-loop, along side the state variables. The work in [31] employs the backstepping active disturbance rejection control. A sliding mode based extended disturbance observer is designed in [30]. The work presented here employs LESO based linear control [24,25] for designing the robust controller. The resulting controller design follows the principle of ADRC in linear set-up.

The main contribution of this paper is to present the aerodynamic model and control, based on ESO technique for the robust control of the CRUAV for trajectory tracking tasks subject to unknown uncertainties and total disturbances. The mathematical model of the coaxial rotorcraft is obtained through Newton–Euler formalism. In general, the ADRC technique is not applicable to underactuated systems. However, as shown in [14] a simplified model commonly adopted for helicopters is feedback linearizable by taking a dynamic extension of the thrust actuation and hence stabilizable by means of ADRC. This would also lead to more complex control equations. Moreover, the aerodynamic forces acting on the planar axis cannot be estimates. To solve this problem, a hierarchical flight control is proposed due to the cascade property between translational and rotational subsystems coming up with a high level position control and a low level attitude control and separate the control design into an inner-loop that controls the moments acting on the vehicle, and an outer-loop that controls the thrust force acting on the vehicle [12]. In fact, the thrust vector is effectively oriented in the desired direction by controlling changes to the CRUAV attitude using the inner-loop. The question of modeling the aerodynamic effects is over thrown by considering them as unknown perturbations that act on the system. To address the

issues of uncertainties and external disturbances, an ESO [25] is designed and integrated with the linear controller to achieve robustness. The aim of control action in the proposed methodology is to enhance the performance of conventional governor working in the outer control loop with the observer operating in the inner perturbation reconstruction–rejection loop. The nominal performance for closed-loop system can be recovered if the total uncertainty/disturbance is timely compensated via LESO.

In summary, the results displayed in this paper are original for several reasons:

- Introduction of a control structure that handles unknown variable aerodynamic efforts.
- A hierarchical flight control structure is proposed due to the cascade property between translational and rotational subsystems.
- A LESO based linear control is designed to achieve robustness.
- The convergence of the closed-loop system (with control and observer) is formally established.

The paper is organized as follows. Section 2 briefly describes the dynamical model of the coaxial-rotor UAV. Section 3 introduces the non-linear controller based on LESO integrated with a linear controller and applied to a coaxial-rotor rotorcraft. Simulations results are presented in Section 4 to show the behavior and stability of the closed-loop system, while conclusions and future directions will be explained in Section 5.

2. Vehicle dynamics modeling

The mathematical model for the flying machines [1,2] such as the coaxial-rotor MAV is essential for designing a control algorithm with satisfactory performance. In this section, we describe the nonlinear model for coaxial-rotor system in case of quasi-stationary flight as depicted in Fig. 1. This system is an underactuated mechanical system with six degrees of freedom, but only four degrees of freedom can be controlled independently with four control inputs, which are the thrust T and the three control torque $\Gamma = (\tau_\phi, \tau_\theta, \tau_\psi)^T$ produced by both rotors and Swashplate incidence angles. The main thrust is used to compensate the gravity force and to control the vertical movement. The horizontal movements are controlled by directing the force vector in the appropriate direction (thrust vectoring control) through the cyclic swashplate. Control moments are used to control the aircraft body orientation which controls the rotorcraft horizontal movement.

Consider the coaxial-rotor depicted in Fig. 2 as a solid body incorporating a force and moment generation process [3,6]. Let $\mathcal{B} := (G, x_b, y_b, z_b)$ be the body-fixed frame attached to the center of gravity of the aerial vehicle, where x_b is the longitudinal axis, y_b is the lateral axis and z_b is the vertical direction in hover conditions and $\mathcal{I} := (O, x_I, y_I, z_I)$ is the Earth frame. The generalized coordinates describing the rotorcraft position and orientation are $q = [\xi, \eta]^T$, where $\xi = (x, y, z)^T \in \mathbb{R}^3$ represents the translation coordinates relative to the inertial frame and $\eta = (\phi, \theta, \psi)^T \in \mathbb{R}^3$ are the Euler angles representing the orientation of the rotorcraft in frame $\mathcal{I}$, where ϕ is the Roll angle (rotation around the x -axis), θ is the Pitch angle (rotation around the y -axis), and ψ the Yaw angle (rotation around the z -axis). $\mathcal{R}_\eta = \mathcal{R}_\psi \mathcal{R}_\theta \mathcal{R}_\phi : \mathcal{B} \rightarrow \mathcal{I}$ is the rotation matrix between Earth and body coordinate systems given by¹

$$\mathcal{R}_\eta = \begin{pmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{pmatrix} \quad (1)$$

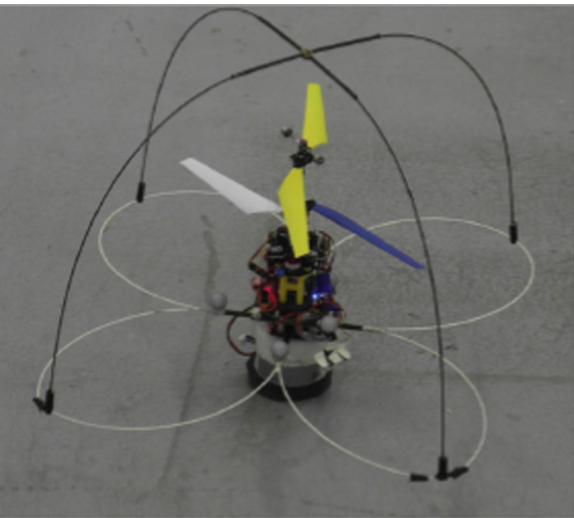


Fig. 1. Prototype of the coaxial rotor [5].

¹ The abbreviations s_* and c_* denote $\sin(\star)$ and $\cos(\star)$, respectively.

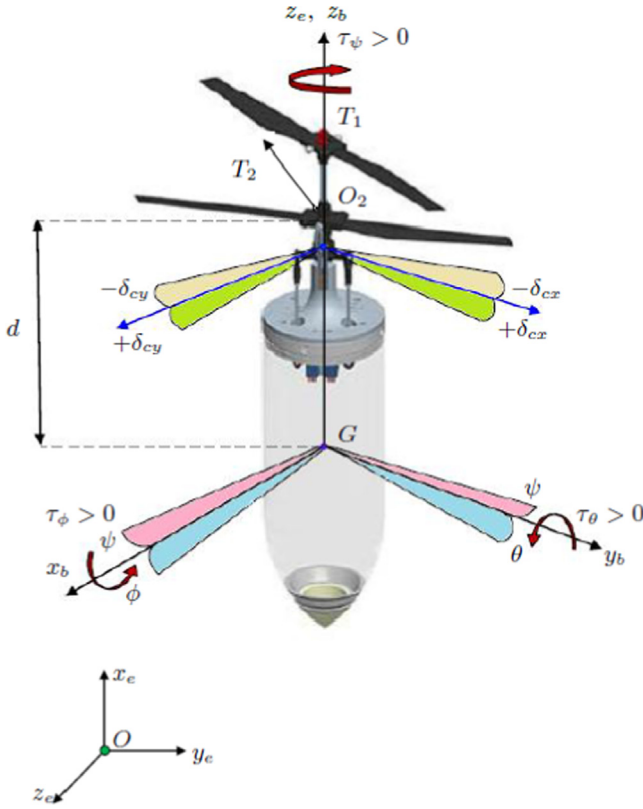


Fig. 2. Coaxial-rotor structure with the associated forces and frames.

The dynamics of a rotorcraft UAV such as the coaxial-rotor helicopter can be represented by the following mathematical model [1,12]:

$$\begin{cases} m\ddot{\xi} &= \mathcal{R}_\eta T - mgz_e + F_{ext} \\ M(\eta)\ddot{\eta} &= \Psi_\eta^T \Gamma - C(\eta, \dot{\eta})\dot{\eta} + \Gamma_{ext} \end{cases} \quad (2)$$

where $M(\eta) = \Psi_\eta^T J \Psi_\eta \in \mathbb{R}^{3 \times 3}$ is an auxiliary positive inertia matrix provided that $(\theta \neq k\pi/2)$, $m \in \mathbb{R}$ specifies the mass, $J \in \mathbb{R}^{3 \times 3}$ is the inertial matrix and g is the gravitational acceleration. T and Γ are, respectively, the aerodynamic forces and moments acting on the coaxial-rotor and $C(\eta, \dot{\eta}) \in \mathbb{R}^{3 \times 3}$ is given by [1]

$$C(\eta, \dot{\eta}) = -\Psi_\eta \ddot{\Psi}_\eta^{-1} \Psi_\eta - \Psi_\eta^T \text{sk}(J \Psi_\eta \dot{\eta}) \Psi_\eta$$

The Euler parameter matrix $\Psi_\eta \in \mathbb{R}^{3 \times 3}$ and the skew anti-symmetric matrix $\text{sk}(\beta) \in \mathbb{R}^{3 \times 3}$ induced by $\beta \in \mathbb{R}^3$ are defined as follows [2,12]:

$$\Psi_\eta = \begin{pmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{pmatrix}; \quad \text{sk}(\beta) = \begin{pmatrix} 0 & -\beta_3 & \beta_2 \\ \beta_3 & 0 & -\beta_1 \\ -\beta_2 & \beta_1 & 0 \end{pmatrix}$$

2.1. Aerodynamics forces and torques generation processes

The coaxial-rotor helicopter includes a mechanism for generating forces and torques that are required to control its motion. In the following, we express the force and torque vectors (T, Γ) generated by both rotors and cyclic swashplate mechanism [6]. The gyroscopic moments induced by tilting the lower rotor and the differential rotors speeds and the drag torque are neglected in the model. This is justified by the fact that the speed differential is small.

2.1.1. Forces generated by the coaxial-rotors

The thrust is the main force generated by the two rotors, which is a function of the rotor angular speed and the cyclic tilt angles. The upper rotor has no swashplate, then, the thrust vector of this rotor always has the same direction, whereas the lower rotor generates both a vertical thrust and lateral forces due to the swashplate incidence angles. Then the thrust vector T is defined [5]:

$$T = \begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix} = \begin{pmatrix} -\kappa_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ -\kappa_\beta \sin \delta_{cx} \Omega_2^2 \\ \kappa_\alpha \Omega_1^2 + \kappa_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2 \end{pmatrix} \quad (3)$$

where $\kappa_\beta > 0$ and $\kappa_\alpha > 0$ are the rotor aerodynamic coefficients, Ω_1 and Ω_2 the rotors rotation speeds, and δ_{cx} and δ_{cy} the swashplate incidence angles.

In addition to the thrust generated by the coaxial-rotor, there exist some other aerodynamic forces acting on the fuselage, which are composed of two kinds of terms. First, the drag forces of the body in the air flow composed of the external wind (disturbance and translational velocities) and the rotor air slipstream whereas the other terms include all the effects coming from geometric parameter uncertainties (mass, inertia, etc.) and unknown perturbation acting on the system (wind gusts). Let us denote by F_{ext} all the aerodynamic efforts viewed as an unknown perturbation acting on the system in the inertial frame.

2.1.2. Moments induced by the coaxial-rotor

The thrust vector T generates the torque vector due to the separation between the center of the mass G and the rotor hubs. The gravitational force does not generate a torque since the vehicle is free to rotate around its centre of mass. According to [3], the torque vector Γ applied to the airframe by the thrust vector is defined by

$$\Gamma = \begin{pmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{pmatrix} = \begin{pmatrix} -d\kappa_\beta \sin \delta_{cx} \Omega_2^2 \\ d\kappa_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ \gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2 \end{pmatrix} \quad (4)$$

with d being the distance between the centre of gravity and the lower rotor center of rotation and $\gamma_1 > 0$ and $\gamma_2 > 0$ are the yaw aerodynamic coefficients.

Additionally, the moment $\Gamma_{ext} = l_{ext} \wedge F_{ext}$ resulting from the aerodynamic effects with a lever arm l_{ext} .

According to (4), the force vector in (3) can be rewritten as follows [3]:

$$T = (0, 0, T_z)^T + \Sigma \Gamma \quad \text{with} \quad \Sigma = \frac{1}{d} \text{sk}(z_e) \quad (5)$$

The matrix Σ displays the coupling between the mechanism force and moment generation which produces small parasitic forces named small body forces, and introduces zero-dynamics into the model of the system [5,14].

2.2. Complete model

By merging (3) and (4) in a Newton Euler model (2), a theoretical non-linear model of a coaxial helicopter for a general 3D flight mode is obtained as [3]

$$\begin{pmatrix} \dot{\xi} \\ m\dot{v} \\ \dot{\eta} \\ M(\eta)\dot{w} \end{pmatrix} = \begin{pmatrix} v \\ T_z R_\eta z_e - mgz_e + \mathcal{R}_\eta \Sigma \Gamma + F_{ext} \\ w \\ \Psi_\eta^T \Gamma - C(\eta, \dot{\eta})w + \Gamma_{ext} \end{pmatrix} \quad (6)$$

where $v = (v_x, v_y, v_z)^T$ and $w = (w_\phi, w_\theta, w_\psi)^T$ are the linear and angular velocities in $\mathcal{I}$, respectively. For the development of the control laws, the following assumptions are needed:

Assumption 1. It is assumed that the pitch and roll angles satisfy the following inequalities:

$$-\pi/2 \leq \phi(t) \leq \pi/2, \quad -\pi/2 \leq \theta(t) \leq \pi/2$$

Assumption 2. The reference trajectories and their first and second time derivatives are assumed to be bounded.

Remark 1. From a practical point of view, the model may be simplified knowing that the lateral tilting (δ_{cx}, δ_{cy}) are imposed to be small [5]. Therefore, we can consider that ($\cos \delta_{ci} \approx 1$) and ($\sin \delta_{ci} \approx \delta_{ci}$). Consequently, from the expression (3) and (4), the two lateral forces (T_x, T_y) due to small cyclic swashplate incidence angles (δ_{cx}, δ_{cy}) are small with respect to the vertical thrust and consequently ($\Sigma \Gamma \approx 0$).

It is important to note that the change of variables in (3) and (4) under Remark 1 defines a diffeomorphism. In other words, the

direction by controlling changes to the UAV attitude using the inner-loop. The coaxial-rotor model is separated into two connected subsystems by decoupling the translational dynamics and rotational dynamics. The rotation dynamics do not depend on translation components, while the translational dynamics depend on angles via the rotation matrix $\mathcal{R}_\eta$. Consequently, the coaxial-rotor model (6) can be written as follows:

$$\begin{aligned} \Sigma_T : \begin{cases} \dot{\xi} &= v \\ m\dot{v} &= T_z \mathcal{R}_\eta^d z_e + T_z h(\eta_d, \delta_\eta) - mgz_e + F_{ext} \end{cases} \\ \Sigma_R : \begin{cases} \dot{\eta} &= w \\ M(\eta)\dot{w} &= \Psi_\eta^T \Gamma - C(\eta, w)w + \Gamma_{ext} \end{cases} \end{aligned} \quad (8)$$

The vector $h(\eta_d, \delta_\eta)$ represents the interconnection term between the translation and rotation dynamics. The components of the interconnection term $h = (h_x \ h_y \ h_z)^T$ are given by

$$\begin{aligned} \begin{cases} h_x &= [h_2(\cdot) \cos \phi_d - h_1(\cdot) \sin \theta_d - h_1(\cdot)h_2(\cdot)] [\cos \psi_d - h_3(\cdot)] - h_3(\cdot) \cos \phi_d \sin \theta_d + \bar{h}_3(\cdot) \sin \phi_d + \bar{h}_1(\cdot) [\bar{h}_3(\cdot) + \sin(\psi_d)] \\ h_y &= [h_2(\cdot) \cos \phi_d - h_1(\cdot) \sin \theta_d - h_1(\cdot)h_2(\cdot)] [\sin \psi_d + \bar{h}_3(\cdot)] + \bar{h}_3(\cdot) \cos \phi_d \sin \theta_d + h_3(\cdot) \sin \phi_d + \bar{h}_1(\cdot) [h_3(\cdot) - \cos(\psi_d)] \\ h_z &= h_2 h_3 + \cos \phi_d h_2 + \cos \theta_d h_3. \end{cases} \\ \text{with} \\ \begin{cases} h_1 &= 2 \sin(\delta_\phi/2) \sin(\phi_d + \delta_\phi/2), \quad \bar{h}_1 = 2 \sin(\delta_\phi/2) \cos(\phi_d + \delta_\phi/2) \\ h_2 &= 2 \sin(\delta_\theta/2) \cos(\theta_d + \delta_\theta/2), \quad \bar{h}_2 = 2 \sin(\delta_\theta/2) \sin(\theta_d + \delta_\theta/2) \\ h_3 &= 2 \sin(\delta_\psi/2) \sin(\psi_d + \delta_\psi/2), \quad \bar{h}_3 = 2 \sin(\delta_\psi/2) \cos(\psi_d + \delta_\psi/2) \end{cases} \end{aligned} \quad (9)$$

original control inputs ($\Omega_1, \Omega_2, \delta_{cx}, \delta_{cy}$) can be recovered from (T_z, Γ) by applying the inverse transformation, which are given by

$$\Omega_1^2 = \frac{\gamma_2 T_z - \kappa_\beta \tau_\psi}{\kappa_\alpha \gamma_2 + \kappa_\beta \gamma_1}, \quad \Omega_2^2 = \frac{\gamma_1 T_z - \kappa_\alpha \tau_\psi}{\kappa_\beta \gamma_1 + \kappa_\alpha \gamma_2}, \quad \delta_{cx} = -\frac{\tau_\phi}{d\kappa_\beta \Omega_2^2}, \quad \delta_{cy} = \frac{\tau_\theta}{d\kappa_\beta \Omega_2^2} \quad (7)$$

3. Control design

A hierarchical flight control approach based on the Extended State Observer technique is proposed to solve the path tracking problem for a small coaxial helicopter in the presence of unknown aerodynamic efforts. The resulting nonlinear controller is thus, simple, robust and easy to implement. The aim of control action in the proposed methodology is to enhance the performance of conventional controller working in the outer control loop with the Extended State Observer operating in the inner perturbation reconstruction–rejection loop. The basic idea for designing a hierarchical flight controller is to split the controller into an inner-loop that controls the moments acting on the aircraft, and an outer-loop that controls the thrust force acting on the aircraft. In fact, the thrust vector is effectively oriented in the desired

direction by controlling changes to the UAV attitude using the inner-loop. The coaxial-rotor model is separated into two connected subsystems by decoupling the translational dynamics and rotational dynamics. The rotation dynamics do not depend on translation components, while the translational dynamics depend on angles via the rotation matrix $\mathcal{R}_\eta$. Consequently, the coaxial-rotor model (6) can be written as follows:

$$\Gamma = J \Psi_\eta \tilde{\Gamma} + \Phi_\eta^T C(\eta, w)w \quad (10)$$

where $\Phi_\eta = \Psi_\eta^{-1}$ and $\tilde{\Gamma} = (\tilde{\tau}_\phi, \tilde{\tau}_\theta, \tilde{\tau}_\psi)^T \in \mathbb{R}^3$ are considered in the sequel as new control inputs. Let us take $\tilde{\Gamma}_{ext} = M(\eta)^{-1} \Gamma_{ext}$. Therefore, we can rewrite the model (8) as follows:

$$\begin{aligned} \Sigma_T : \begin{cases} \dot{\xi} &= v \\ m\dot{v} &= T_z \mathcal{R}_\eta^d z_e + T_z h(\eta_d, \delta_\eta) - mgz_e + F_{ext} \end{cases} \\ \Sigma_R : \begin{cases} \dot{\eta} &= w \\ \dot{w} &= \tilde{\Gamma} + \tilde{\Gamma}_{ext} \end{cases} \end{aligned} \quad (11)$$

The true inputs of the system are T_z and Γ , all other quantities ($\mu_{xi}, \tilde{\Gamma}$) are intermediate controls. The thrust T_z and the new command torque $\tilde{\Gamma}$ are the new control inputs for the system (11). The thrust T_z and the rotation matrix $\mathcal{R}_\eta^d$ are responsible for the translational movements. The rotation matrix $\mathcal{R}_\eta$ in terms of Euler angles can also be regarded as the output of the rotational subsystem. This

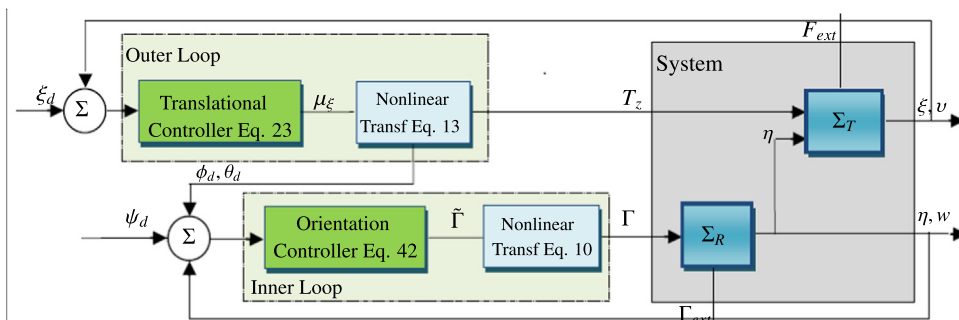


Fig. 3. Block diagram of cascade systems.

can be considered as the cascade property of the coaxial-rotor system (11). Consequently, translational and rotational controllers can be designed separately as depicted in Fig. 3. The translational controller is firstly designed to extract the desired thrust T_z^d and attitude matrix $\mathcal{R}_\eta^d$. T_z is considered as the desired value T_z^d . The desired attitude information can enable a rotorcraft to track the desired trajectory.

From the system Σ_T in (11), let us define a virtual control vector $\mu_\xi = (\mu_x \mu_y \mu_z)^T \in \mathbb{R}^3$ as follows [12]:

$$\mu_\xi = q(T_z, \phi_d, \theta_d) = \frac{1}{m} T_z \mathcal{R}_\eta^d z_e - g z_e \quad (12)$$

where $q(\cdot) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a continuous invertible function and (ϕ_d, θ_d) is the desired Roll and Pitch angles. According to (12), we can calculate the desired force vector magnitude and desired $\mathcal{R}_\eta^d$ as follows [12]:

$$\begin{cases} T_z &= m \sqrt{\mu_x^2 + \mu_y^2 + (\mu_z + g)^2} \\ \phi_d &= \sin^{-1} \left[\frac{m}{T_z} (\mu_x \sin \psi_d - \mu_y \cos \psi_d) \right] \\ \theta_d &= \sin^{-1} \left[\frac{m}{T_z \cos \phi_d} (\mu_x \cos \psi_d + \mu_y \sin \psi_d) \right] \end{cases} \quad (13)$$

We define the following error vectors: $\delta_2 = [\delta_\eta \ \delta_w]^T \in \mathbb{R}^6$ with $\delta_\eta = [\delta_\phi \ \delta_\theta \ \delta_\psi]^T = \eta - \eta_d$, and $\delta_w = w - \dot{\eta}_d$. The desired trajectory of the attitude subsystem η_d contains the desired Euler angles, which are ϕ_d and θ_d given by (13) and ψ_d is a reference trajectory chosen by the Pilot or by a high-level navigation system. Defining the tracking errors $\delta_1 = [\delta_\xi \ \delta_v]^T = [\xi - \xi_d \ v - \dot{\xi}_d]^T \in \mathbb{R}^6$, the two subsystems (11) can be written as

$$\begin{cases} \dot{\delta}_1 &= A_1 \delta_1 + B_1 \left[\mu_\xi - \ddot{\xi}_d + \frac{1}{m} [T_z h(\eta_d, \delta_\eta) + F_{ext}] \right] \\ \dot{\delta}_2 &= A_2 \delta_2 + B_2 [\ddot{\eta}_d - \ddot{\eta}_d + \ddot{\eta}_d] \end{cases} \quad (14)$$

The control objective is then to find control laws T_z and $\ddot{\eta}_d$ such that the tracking errors δ_1 and δ_2 converge asymptotically to zero. Let $\Delta(h, F_{ext}) = \frac{1}{m} [T_z h(\eta_d, \delta_\eta) + F_{ext}]$ be the interconnection term and the total disturbance. According to Assumptions 1 and 2, one can easily show that the interconnection term $h(\eta_d, \delta_\eta)$ is bounded. The matrices $A_1 \in \mathbb{R}^{6 \times 6}$, $B_1 \in \mathbb{R}^{6 \times 3}$, $A_2 \in \mathbb{R}^{6 \times 6}$, and $B_2 \in \mathbb{R}^{6 \times 3}$ are defined as

$$A_1 = A_2 = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}, \quad B_1 = B_2 = \begin{bmatrix} 0 \\ I \end{bmatrix} \quad (15)$$

where I is an identity matrix of dimension 3×3 .

The application of the conventional active disturbance rejection control to the rotorcraft UAV would require taking a dynamic extension of the control input T_z [14]. In other words, the system input T_z and its derivative $\dot{T}_z$ would then be viewed as internal states of the controller. This would also lead to more complex control equations than the ones that are presented in this paper. Moreover, the aerodynamic forces acting on the planar axis cannot be estimates through the ESO.

3.1. Design of the position controller

Let ξ_d be the reference trajectory of the translational position ξ . By merging (12) and Σ_T in (11), the translation subsystem in (11) becomes a linear form (without $\Delta(h, F_{ext})$) with three dimensional double-integrator. Generally, the command μ_ξ involving a linear combination of the state δ_1 is:

$$\mu_\xi = \ddot{\xi}_d - K_\xi \delta_\xi - K_v \delta_v + u_{oc} \quad (16)$$

where $K_\xi = \text{diag}[k_x \ k_y \ k_z]$ and $K_v = \text{diag}[k_{v_x} \ k_{v_y} \ k_{v_z}]$ are the controller gains which need to be chosen so as to satisfy the desired

performances specifications. $u_{oc} \in \mathbb{R}^3$ is the compensation term of the $\Delta(h, F_{ext})$.

We define

$$K_1 = \begin{bmatrix} k_x & 0 & 0 & k_{v_x} & 0 & 0 \\ 0 & k_y & 0 & 0 & k_{v_y} & 0 \\ 0 & 0 & k_z & 0 & 0 & k_{v_z} \end{bmatrix} \quad (17)$$

The controller parameters, k_i and k_{v_i} , with $i = x, y, z$ are selected such that the matrix $A_{1_H} = A_1 - B_1 K_1$ is stable.

Remark 2. In practical applications, the parameters of the position controller k_i and k_{v_i} can be determined by pole placement method.

By substituting (16) into (14), we get

$$\dot{\delta}_1 = A_{1_H} \delta_1 + B_1 [u_{oc} + \Delta(h, F_{ext})] \quad (18)$$

The term $\Delta(h, F_{ext})$ will be compensated using a linear Extended State Observer to ensure the robustness of the control strategy. If A_{1_H} is Hurwitz, the subsystem Σ_T (without disturbance $\Delta(h, F_{ext})$) is globally asymptotically stable.

3.1.1. Linear Extended State Observer

In this section, a Linear Extended State Observer is designed to estimate the unknown total disturbance. Define $\varsigma = [\varsigma_1, \varsigma_2, \varsigma_3]^T = [\xi, v, \Delta(h, F_{ext})]^T$. To construct this observer, the uncertain system Σ_T in (11) is augmented as

$$\begin{cases} \dot{\varsigma}_1 &= \varsigma_2 \\ \dot{\varsigma}_2 &= \varsigma_3 + \mu_\xi \\ \dot{\varsigma}_3 &= p(\varsigma) \\ y &= \varsigma_1 \end{cases} \quad (19)$$

Here $p(\varsigma) = \dot{\Delta}(h, F_{ext})$ is assumed to be unknown, but bounded function. The augmented system (19) can be written as

$$\dot{\varsigma} = A_0 \varsigma + B_0 \mu_\xi + E_0 p(\varsigma) \quad (20)$$

The matrices $A_0 \in \mathbb{R}^{9 \times 9}$, $B_0 \in \mathbb{R}^{9 \times 3}$ and $E_0 \in \mathbb{R}^{9 \times 3}$ are defined as

$$A_0 = \begin{bmatrix} 0 & I & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 0 \\ I \\ 0 \end{bmatrix}, \quad E_0 = \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix}$$

and $C = [I \ 0 \ 0] \in \mathbb{R}^{3 \times 9}$. The LESO (with the function $\mathcal{G}_i(\cdot) = \alpha_i$) for the system (19) is given by

$$\begin{cases} \dot{\hat{\varsigma}}_1 &= \hat{\varsigma}_2 + \frac{\alpha_1}{\epsilon} I (\varsigma_1 - \hat{\varsigma}_1) \\ \dot{\hat{\varsigma}}_2 &= \mu_\xi + \hat{\varsigma}_3 + \frac{\alpha_2}{\epsilon^2} I (\varsigma_1 - \hat{\varsigma}_1) \\ \dot{\hat{\varsigma}}_3 &= \frac{\alpha_3}{\epsilon^3} I (\varsigma_1 - \hat{\varsigma}_1) \end{cases} \quad (21)$$

where α_j and ϵ are the observer gains. Define $\hat{\varsigma} = [\hat{\varsigma}_1 \ \hat{\varsigma}_2 \ \hat{\varsigma}_3]^T$, the state space model of the LESO dynamics (21) can be written as

$$\dot{\hat{\varsigma}} = A_0 \hat{\varsigma} + B_0 \mu_\xi + L C (\varsigma - \hat{\varsigma}) \quad (22)$$

where $L = [\frac{\alpha_1}{\epsilon} I \ \frac{\alpha_2}{\epsilon^2} I \ \frac{\alpha_3}{\epsilon^3} I]^T \in \mathbb{R}^{9 \times 3}$ is the observer gain vector.

Remark 3. The LESO is employed instead of the Nonlinear Extended State Observer – NESO – as the use of linear ESO offers certain advantages. Firstly, the observer gains can be systematically chosen through pole placement, whereas the various parameters appearing in NESO are usually chosen by trial and error procedure. Secondly the closed loop stability for LESO can be

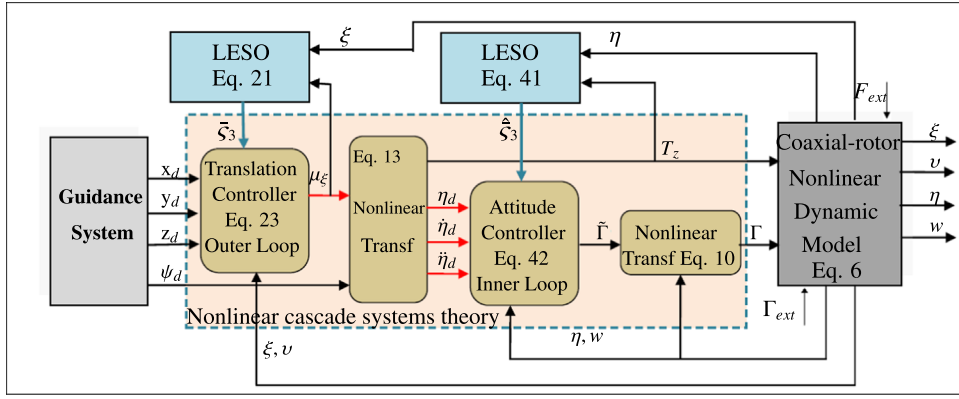


Fig. 4. Structure of the inner-outer-loop nonlinear controller.

established conclusively as shown in the next section. Lastly, the LESO is easy from the hardware implementation point of view.

3.1.2. LESO based controller

Using the estimate in (16), the resulting controller takes the form as

$$\begin{aligned}\mu_{\xi} &= \ddot{\xi}_d - K_{\xi}(\hat{\xi}_1 - \xi_d) - K_{v_{\xi}}(\hat{\xi}_2 - \dot{\xi}_d) + u_{oc} \\ u_{oc} &= -\hat{\xi}_3\end{aligned}\quad (23)$$

where $\hat{\xi}_3$ is the estimate of the total disturbance acting on the system. A functional block diagram of the ESO based controller is shown in Fig. 4.

3.2. Closed loop stability

In this section, results on closed stability [21,22] are presented. Firstly the observation error can be written as

$$\tilde{\zeta}_i = \zeta_i - \hat{\zeta}_i \quad (24)$$

Let us define

$$q_i(t) = \frac{\tilde{\zeta}_i(et)}{e^{n+1-i}} \quad (25)$$

with $n = 2$ and $i = 1, \dots, n+1$.

From (20), (22) and (25), the observer error dynamics can be also rewritten as

$$\begin{aligned}\dot{q}_1 &= q_2 - \alpha_1 I q_1 \\ \dot{q}_2 &= q_3 - \alpha_2 I q_1 \\ \dot{q}_3 &= \epsilon p(\zeta) - \alpha_3 I q_1\end{aligned}\quad (26)$$

Define $q = [q_1 \ q_2 \ q_3]^T$, the observer error dynamics (26) can be written as

$$\dot{q} = (A_0 - L_x C)q + E_0 \epsilon p(\zeta) \quad (27)$$

where $L_x = [\alpha_1 I \ \alpha_2 I \ \alpha_3 I]^T \in \mathbb{R}^{9 \times 3}$ and

$$LC = \begin{bmatrix} \frac{\alpha_1}{\epsilon} I & 0 & 0 \\ \frac{\alpha_2}{\epsilon^2} I & 0 & 0 \\ \frac{\alpha_3}{\epsilon^3} I & 0 & 0 \end{bmatrix}; \quad L_x C = \begin{bmatrix} \alpha_1 I & 0 & 0 \\ \alpha_2 I & 0 & 0 \\ \alpha_3 I & 0 & 0 \end{bmatrix}$$

Consider the following expression:

$$\dot{\delta}_1 = A_1 \delta_1 + B_1 [\mu_{\xi} - \ddot{\xi}_d + \Delta(h, F_{ext})] \quad (28)$$

The control law (23) can be rewritten as

$$\mu_{\xi} = \ddot{\xi}_d - K_1 \delta_1 + [K_1 \ 0_{3 \times 3}] \tilde{\zeta} - \hat{\xi}_3 \quad (29)$$

Substituting (29) into (28), we get

$$\dot{\delta}_1 = A_{1H} \delta_1 + [B_1 K_1 \ B_1] \tilde{\zeta} \quad (30)$$

Combining (27) and (30), we get

$$\begin{bmatrix} \dot{\delta}_1 \\ \dot{q} \end{bmatrix} = \begin{bmatrix} A_{1H} & [B_1 K_1 \ B_1] \epsilon^{3-i} \\ 0 & A_0 - L_x C \end{bmatrix} \begin{bmatrix} \delta_1 \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ \epsilon E_0 \end{bmatrix} p(\zeta) \quad (31)$$

For stability analysis results, the following assumption is made:

Assumption 3. The rate of change of disturbance $p(\zeta)$ is assumed bounded.

Under Assumption 3, the results on observer error dynamics and closed loop system stability are presented as follows.

3.2.1. Observer error dynamics

Theorem 1. Consider the observer error dynamics of (27) under Assumption 3, if the observer gain vector L_x is chosen such that $(A_0 - L_x C)$ is stable, then q exponentially converges to the bounded ball $B_{r_1} = q \in \mathbb{R}^9$, $\| \tilde{\zeta} \| \leq 2\epsilon \lambda_{\max}(P_0) h_{\max}$, where $\lambda_{\max}(P_0)$ is the maximum eigenvalue of P_0 , and P_0 solution of the equation $(A_0 - L_x C)^T P_0 + P_0 (A_0 - L_x C) = -I_{9 \times 9}$, and $h_{\max}$ is the absolute maximum value of $p(\zeta)$.

Proof. Define the Lyapunov candidate function V_0 as

$$V_0 = q^T P_0 q \quad (32)$$

Taking the derivative of V_0 along the trajectory, we find

$$\begin{aligned}\dot{V}_0 &= q^T [(A_0 - L_x C)^T P_0 + P_0 (A_0 - L_x C)] q + 2\epsilon q^T P_0 E_0 p(\zeta) \\ &\leq -\|q\|^2 + 2\epsilon \|q\| \|P_0\| h_{\max} \leq -\|q\|^2 \left[1 - \frac{2\epsilon \lambda_{\max}(P_0) h_{\max}}{\|q\|} \right]\end{aligned}\quad (33)$$

Thus $\dot{V}_0 < 0$ whenever $\|q\| > 2\epsilon \lambda_{\max}(P_0) h_{\max}$. Thus, q exponentially converges to the bounded ball $B_{r_1} = q \in \mathbb{R}^9 \mid \|q\| \leq 2\epsilon \lambda_{\max}(P_0) h_{\max}$. We can conclude that, when ϵ is very small, then the bounded ball B_{r_1} is very small and consequently q is also very small.

Remark 4. From (25), the convergence of q to small ball B_{r_1} leads to the convergence of $\tilde{\zeta}$ to a smaller ball.

3.2.2. Closed loop stability

Next, consider the closed loop dynamics given by (31). Define $\delta_{cl} = [\delta_1 \ q]^T$, the closed loop dynamics (31) can be rewritten as

$$\dot{\delta}_{cl} = A_{cl} \delta_{cl} + B_{cl} \epsilon p(\zeta) \quad (34)$$

where

$$A_{cl} = \begin{bmatrix} A_{1H} & [B_1 K_1 \ B_1] \epsilon^{3-i} \\ 0 & A_0 - L_x C \end{bmatrix}, \quad B_{cl} = \begin{bmatrix} 0 \\ E_0 \end{bmatrix}$$

Theorem 2. Consider the closed loop system dynamics of (34) under Assumption 3, if the gain vector K_1 and the observer gain vector L_x are chosen such that A_{cl} is Hurwitz, then $\delta_{cl} = [\delta_1 \ q]^T$ exponentially converges to the bounded ball $B_{r_2} = \delta_{cl} \in \mathbb{R}^{15}$, $\|\delta_{cl}\| \leq 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$, where $\lambda_{\max}(P_{cl})$ is the maximum eigenvalue of P_{cl} and P_{cl} solution of the equation $A_{cl}^T P_{cl} + P_{cl} A_{cl} = -I$, and $h_{\max}$ is the absolute maximum value of $p(\zeta)$.

Proof. Define the Lyapunov candidate function V_{cl} as

$$V_{cl} = \delta_{cl}^T P_{cl} \delta_{cl} \quad (35)$$

The derivative of V_{cl} with respect to time is

$$\begin{aligned} \dot{V}_{cl} &= \delta_{cl}^T [A_{cl}^T P_{cl} + P_{cl} A_{cl}] \delta_{cl} + 2\epsilon \delta_{cl}^T P_{cl} B_{cl} p(\zeta) \leq \\ &- \|\delta_{cl}\|^2 + 2\epsilon \|\delta_{cl}\| \|P_{cl}\| h_{\max} \leq -\|\delta_{cl}\| (\|\delta_{cl}\| - 2\epsilon\lambda_{\max}(P_{cl})h_{\max}) \end{aligned} \quad (36)$$

Thus $\dot{V}_{cl} < 0$ whenever $\|\delta_{cl}\| > 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$. Thus, δ_{cl} exponentially converges to the bounded ball $B_{r_2} = \delta_{cl} \in \mathbb{R}^{15}$, $\|\delta_{cl}\| \leq 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$. We can conclude that when ϵ is very small, then the bounded ball B_{r_2} is very small and consequently δ_{cl} is also very small.

From (34), it can be seen that the eigenvalues of the error dynamics are given by the eigenvalues of A_{1H} and $(A_0 - L_x C)$. Stability of the error dynamics (34) can always be ensured by placing the controller and observer poles appropriately. Further, since (34) is stable, it is obvious that under the assumption of boundedness of $p(\zeta)$, the bounded input-bounded output stability for the system (34) is assured. A special case arises when $p(\zeta) = 0$, i.e., when the rate of change of uncertainty $\Delta(h, F_{ext})$ is zero. In this case, the error dynamics (34) is asymptotically stable. From (26), it can be observed that the LESO offers asymptotic convergence of state estimate for constant disturbance case, i.e., for the case when rate of change of total disturbance is zero.

One approach to achieve asymptotic convergence for LESO error dynamics for the class of disturbances that are time-varying but can be expressed in time polynomial function form such as $\Delta(t) = d_0 + d_1 t + d_2 t^2 + \dots$ is to use the generalized or higher order ESO [29]. With the generalized ESO, it can be shown that if the r th order derivative of disturbance is zero, then by designing an r th order ESO, i.e., by considering r extended states, asymptotic convergence for the LESO state estimate can be achieved.

3.3. Design of the attitude controller

Consider the system Σ_R in (11) which must track the desired orientation $\eta_d = (\phi_d, \theta_d, \psi_d)^T$. ψ_d represents the yaw reference trajectory generated by the Pilot or high-level guidance, the two desired trajectories $(\phi_d, \theta_d)^T$ are obtained by the expression (13). The attitude subsystem Σ_R (11) is represented as a linear form (without $\tilde{F}_{ext}$) with three dimensional double-integrator. The command $\tilde{F}$ involving a linear combination of the state δ_2 is

$$\tilde{F} = \ddot{\eta}_d - K_\eta(\eta - \eta_d) - K_w(w - \dot{\eta}_d) + \Gamma_{oc} \quad (37)$$

where $K_\eta = \text{diag}[k_\phi \ k_\theta \ k_\psi]$ and $K_w = \text{diag}[k_{w_\phi} \ k_{w_\theta} \ k_{w_\psi}]$ are the controller gains which need to be chosen so as to satisfy the desired performances specifications. $\Gamma_{oc} \in \mathbb{R}^3$ is the compensation term of the $\tilde{F}_{ext}$ assumed bounded.

We define

$$K_2 = \begin{bmatrix} k_\phi & 0 & 0 & k_{w_\phi} & 0 & 0 \\ 0 & k_\theta & 0 & 0 & k_{w_\theta} & 0 \\ 0 & 0 & k_\psi & 0 & 0 & k_{w_\psi} \end{bmatrix} \quad (38)$$

The controller parameters, k_η and k_w are selected such that the matrix $A_{2H} = A_2 - B_2 K_2$ is Hurwitz. Substituting (37) into (14), we get

$$\dot{\delta}_2 = A_{2H} \delta_2 + B_2(\Gamma_{oc} + \Gamma_{ext}) \quad (39)$$

The disturbance term Γ_{ext} will be compensated using a Linear Extended State Observer to ensure the robustness of the control strategy. If A_{2H} is Hurwitz, the subsystem Σ_R (without disturbance term Γ_{ext}) is globally asymptotically stable.

3.3.1. Linear Extended State Observer

In this section, a Linear Extended State Observer is designed to estimate the unknown total disturbance in (11). Define $\bar{\zeta} = [\bar{\zeta}_1, \bar{\zeta}_2, \bar{\zeta}_3]^T = [\eta, w, \Gamma_{ext}]^T$. To construct this observer, the uncertain system (11) is augmented as

$$\dot{\bar{\zeta}} = A_0 \bar{\zeta} + B_0 \tilde{F} + E_0 \Delta(\bar{\zeta}) \quad (40)$$

Here, $\Delta(\bar{\zeta}) = \tilde{F}_{ext}$ is assumed to be unknown, but bounded function. Define $\hat{\bar{\zeta}} = [\hat{\bar{\zeta}}_1, \hat{\bar{\zeta}}_2, \hat{\bar{\zeta}}_3]^T$, the LESO (with the function $\mathcal{G}_i(\cdot) = \alpha_i$) for the system (40) is given by

$$\dot{\hat{\bar{\zeta}}} = A_0 \hat{\bar{\zeta}} + B_0 \tilde{F} + LC(\bar{\zeta} - \hat{\bar{\zeta}}) \quad (41)$$

where α_j and ϵ are the observer gains.

3.3.2. LESO based controller

Using the estimate in (37), the resulting controller takes the form as

$$\begin{cases} \tilde{F} &= \ddot{\eta}_d - K_\eta(\hat{\bar{\zeta}}_1 - \eta_d) - K_w(\hat{\bar{\zeta}}_2 - \dot{\eta}_d) + \Gamma_{oc} \\ \Gamma_{oc} &= -\hat{\bar{\zeta}}_3 \end{cases} \quad (42)$$

In the control law (42), the last term Γ_{oc} is added to compensate the total disturbance term acting on the model. By substituting (42) into Σ_R (14), the closed-loop system is given by

$$\dot{\delta}_2 = A_{2H} \delta_2 + [B_2 K_2 \ K_2] \hat{\bar{\zeta}} \quad (43)$$

3.4. Closed loop stability

Firstly the observation error can be written as

$$\tilde{\zeta}_i = \bar{\zeta}_i - \hat{\zeta}_i \quad (44)$$

Let us define

$$\bar{Q}_i(t) = \frac{\tilde{\zeta}_i(e t)}{e^{n+1-i}} \quad (45)$$

with $n = 2$ and $i = 1, \dots, n+1$. Define $\bar{Q} = [\bar{Q}_1 \ \bar{Q}_2 \ \bar{Q}_3]^T$, the observer error dynamics can be written as

$$\dot{\bar{Q}} = (A_0 - L_x C) \bar{Q} + E_0 \epsilon \Delta(\bar{\zeta}) \quad (46)$$

The control law (42) can be rewritten as

$$\tilde{F} = \ddot{\eta}_d - K_2 \delta_2 + [K_2 \ 0_{3 \times 3}] \hat{\bar{\zeta}} - \hat{\bar{\zeta}}_3 \quad (47)$$

Substituting (47) into (43), we get

$$\dot{\delta}_2 = A_{2H} \delta_2 + [B_2 K_2 \ B_2] \hat{\bar{\zeta}} \quad (48)$$

Combining (46) and (48), we get

$$\begin{bmatrix} \dot{\delta}_2 \\ \dot{\bar{Q}} \end{bmatrix} = \begin{bmatrix} A_{2H} & [B_2 K_2 \ B_2] \epsilon^{3-i} \\ 0 & A_0 - L_x C \end{bmatrix} \begin{bmatrix} \delta_2 \\ \bar{Q} \end{bmatrix} + \begin{bmatrix} 0 \\ \epsilon E_0 \end{bmatrix} \Delta(\bar{\zeta}) \quad (49)$$

For stability analysis results, the following assumption is made:

Assumption 4. The rate of change of disturbance $\Delta(\bar{\zeta})$ is assumed bounded.

Under **Assumption 4**, the results on observer error dynamics and closed loop system stability are presented as follows.

3.4.1. Observer error dynamics

Theorem 3. Consider the observer error dynamics of (46) under **Assumption 4**, if the observer gain vector L_x is chosen such that $(A_0 - L_x C)$ is stable, then $\bar{Q}$ exponentially converges to the bounded ball $B_{r_3} = \bar{Q} \in \mathbb{R}^9$, $\|\bar{\zeta}\| \leq 2\epsilon\lambda_{\max}(P_0)h_{\max}$, where $\lambda_{\max}(P_0)$ is the maximum eigenvalue of P_0 , and P_0 solution of the equation $(A_0 - L_x C)^T P_0 + P_0(A_0 - L_x C) = -I_{9 \times 9}$, and $h_{\max}$ is the absolute maximum value of $\Delta(\bar{\zeta})$.

Proof. Define the Lyapunov candidate function V_3 as

$$V_3 = \bar{Q}^T P_0 \bar{Q} \quad (50)$$

Taking the derivative of V_3 along the trajectory, we find

$$\begin{aligned} \dot{V}_3 &= \bar{Q}^T \left[(A_0 - L_x C)^T P_0 + P_0(A_0 - L_x C) \right] \bar{Q} + 2\epsilon \bar{Q}^T P_0 E_0 \Delta(\bar{\zeta}) \leq -\|\bar{Q}\|^2 \\ &\quad + 2\epsilon \|\bar{Q}\| \|P_0\| h_{\max} \leq -\|\bar{Q}\|^2 \left[1 - \frac{2\epsilon\lambda_{\max}(P_0)h_{\max}}{\|\bar{Q}\|} \right] \end{aligned} \quad (51)$$

Thus $\dot{V}_3 < 0$ whenever $\|\bar{Q}\| > 2\epsilon\lambda_{\max}(P_0)h_{\max}$. Thus, $\bar{Q}$ exponentially converges to the bounded ball $B_{r_3} = \bar{Q} \in \mathbb{R}^9$, $\|\bar{Q}\| \leq 2\epsilon\lambda_{\max}(P_0)h_{\max}$. We can conclude that, when ϵ is very small, then the bounded ball B_{r_3} is very small and consequently $\bar{Q}$ is also very small.

Remark 5. From (45), The convergence of $\bar{Q}$ to zero leads to the convergence of $\bar{\zeta}$ to zero and consequently $\hat{\zeta} \rightarrow \bar{\zeta}$.

3.4.2. Closed loop stability

Next, consider the closed loop dynamics given by (49). Define $\bar{\delta}_{cl} = [\delta_2 \ \bar{Q}]^T$, the closed loop dynamics (49) can be rewritten as

$$\dot{\bar{\delta}}_{cl} = A_{cl} \bar{\delta}_{cl} + B_{cl} \epsilon \Delta(\bar{\zeta}) \quad (52)$$

where

$$A_{cl} = \begin{bmatrix} A_{2H} & [B_2 K_2 \ B_2] \epsilon^{3-i} \\ 0 & A_0 - L_x C \end{bmatrix}, \quad B_{cl} = \begin{bmatrix} 0 \\ E_0 \end{bmatrix}$$

Theorem 4. Consider the closed loop system dynamics of (52) under **Assumption 4**, if the gain vector K_2 and the observer gain vector L_x are chosen such that A_{cl} is Hurwitz, then $\bar{\delta}_{cl} = [\delta_2 \ \bar{Q}]^T$ exponentially converges to the bounded ball $B_{r_4} = \bar{\delta}_{cl} \in \mathbb{R}^{15}$, $\|\bar{\delta}_{cl}\| \leq 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$, where $\lambda_{\max}(P_{cl})$ is the maximum eigenvalue of P_{cl} , and P_{cl} solution of the equation $A_{cl}^T P_{cl} + P_{cl} A_{cl} = -I$, and $h_{\max}$ is the absolute maximum value of $\Delta(\bar{\zeta})$.

Proof. Define the Lyapunov candidate function V_4 as

$$V_4 = \bar{\delta}_{cl}^T P_{cl} \bar{\delta}_{cl} \quad (53)$$

The derivative of V_4 with respect to time is

$$\begin{aligned} \dot{V}_4 &= \bar{\delta}_{cl}^T \left[A_{cl}^T P_{cl} + P_{cl} A_{cl} \right] \bar{\delta}_{cl} + 2\epsilon \bar{\delta}_{cl}^T P_{cl} B_{cl} \Delta(\bar{\zeta}) \leq -\|\bar{\delta}_{cl}\|^2 \\ &\quad + 2\epsilon \|\bar{\delta}_{cl}\| \|P_{cl}\| h_{\max} \leq -\|\bar{\delta}_{cl}\| \left(\|\bar{\delta}_{cl}\| - 2\epsilon\lambda_{\max}(P_{cl})h_{\max} \right) \end{aligned} \quad (54)$$

Thus $\dot{V}_4 < 0$ whenever $\|\bar{\delta}_{cl}\| > 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$. Thus, $\bar{\delta}_{cl}$ exponentially converges to the bounded ball $B_{r_4} = \bar{\delta}_{cl} \in \mathbb{R}^{15}$, $\|\bar{\delta}_{cl}\| \leq 2\epsilon\lambda_{\max}(P_{cl})h_{\max}$. We can conclude that when ϵ is very small, then the bounded ball B_{r_4} is very small and consequently $\bar{\delta}_{cl}$ is also very small.

Remark 6. The translational and rotational subsystems are coupled through the interconnection term $h(\eta_d, \delta_\eta)$, which is bounded

and depending on the error term δ_2 as follows [12]:

$$\|h(\eta_d, \delta_\eta)\| \leq k_2 \|\delta_2\|, \quad k_2 > 0$$

Consequently, the fast convergence of the rotational error δ_2 to zero implies the convergence of $h(\eta_d, \delta_\eta)$ to zero and consequently the convergence of the translational error δ_1 to zero. The closed-loop system becomes

$$\begin{cases} \dot{\delta}_1 &= A_{1H} \delta_1 + [B_1 K_1 \ B_1] \epsilon^{3-i} Q \\ \dot{\delta}_2 &= A_{2H} \delta_2 + [B_2 K_2 \ K_2] \epsilon^{3-i} \bar{Q} \end{cases} \quad (55)$$

According to **Theorems 1–4**, the tracking errors $[\delta_1, Q]$ and $[\delta_2, \bar{Q}]$ exponentially converge to small bounded balls B_{r_2} and B_{r_4} , respectively.

Remark 7. The final control inputs (T_z, Γ) (Eqs. (10) and (13)) which are nonlinear and consider system's nonlinearities and coupling between state variables:

$$\begin{aligned} T_z &= m \|\mu_\xi + g z_e\| = m \|\ddot{\xi}_d - K_\xi(\dot{\xi}_1 - \dot{\xi}_d) - K_{v_\xi}(\dot{\xi}_2 - \dot{\xi}_d) - \dot{\xi}_3 + g z_e\| \\ \Gamma &= J \Psi_\eta \tilde{r} + \Phi_\eta^T C(\eta, w) w = J \Psi_\eta [\ddot{\eta}_d - K_\eta(\dot{\xi}_1 - \eta_d) - K_w(\dot{\xi}_2 - \dot{\eta}_d) - \dot{\xi}_3] \\ &\quad + \Phi_\eta^T C(\eta, w) w \end{aligned} \quad (56)$$

4. Simulation results

The proposed control strategy has been tested by simulation in order to check the effectiveness and the performance attained for the path following problem. The coaxial helicopter model parameters used are [1] $m = 0.41$ kg, $g = 9.81$ m s⁻¹, $d = 0.0676$, $J = \text{diag}[1.383 \ 1.383 \ 0.272] \times 10^{-3}$ kg m², $\kappa_\alpha = 3.6835 \cdot 10^{-5}$ N rad⁻² s⁻², $\kappa_\beta = 3.7760 \cdot 10^{-5}$ N rad⁻² s⁻², $\gamma_1 = 1.4765 \cdot 10^{-6}$ N rad⁻² s⁻², $\gamma_2 = 1.3266 \cdot 10^{-6}$ N rad⁻² s⁻². As already noted, the controller task is to make the system position variables $\xi(t)$ and the yaw angle $\psi(t)$ asymptotically track the reference trajectories. The initial conditions for position subsystem of the coaxial rotorcraft are $\xi(0) = [0, 0, 0.5]^T$ m and for rotational subsystem are $\eta(0) = [0, 0, 0.3]^T$ rad. Hence the velocity and the angular velocity are set to 0. In practice, the reference angles $[\phi(t), \theta(t)]$ are processed using second-order low-pass digital filters to reduce noise and to compute both first and second derivatives.

4.1. Case a: flight without aerodynamic forces and moments disturbances

This part involves position and attitude tracking control without taking into consideration the system uncertainties and aerodynamic forces and moments disturbances. Fig. 5 shows the evolution of the vehicle's position and orientation during its flight. It can be seen that even though the commanded position $\{x_d(t), y_d(t), z_d(t), \psi_d(t)\}$ were changed in every moment, the proposed controller is able to drive all these state variables back to the new reference position. Fig. 6 shows the trajectories generated by the lateral and longitudinal controller, i.e., (ϕ, θ) varies to help the vehicle to follow the commands trajectories, ensure the convergence of all state to the reference position and improve the effectiveness of the proposed control strategy. Finally, Fig. 7 illustrates the behavior of the input control applied to each actuator. Indeed, it appears clearly that the obtained input control signals are acceptable and physically realizable.

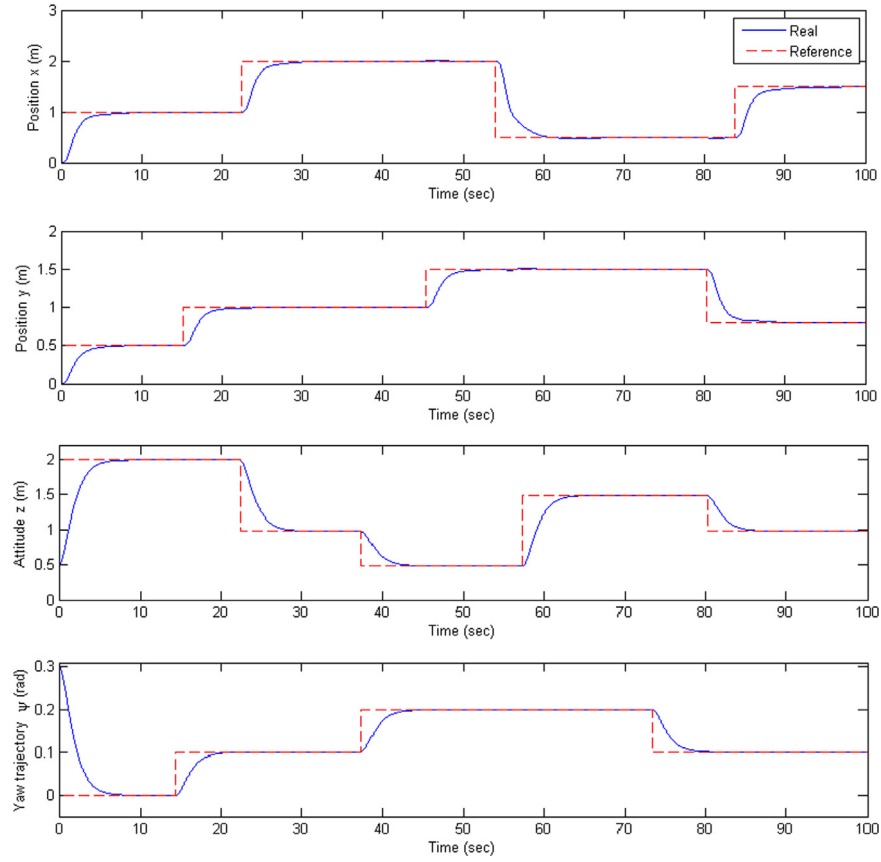


Fig. 5. Case a: position outputs.

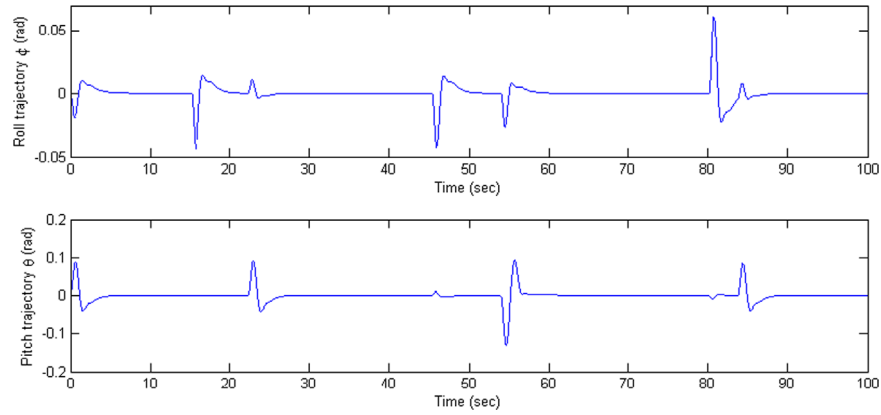


Fig. 6. Case a: roll & pitch trajectories.

4.2. Case b: flight with uncertainty and aerodynamic forces and moments disturbances

The robustness of the proposed controller was tested by considering the aerodynamic forces and moments disturbances. The model uncertainty and aerodynamic forces and moments disturbances on the system are listed below:

- for an uncertainty $\Delta J = 0.5 \text{ J kg m}^2$;

- For the aerodynamic forces occurring at 10, 30 and 40 s, respectively:

$$\frac{1}{m}F_{ext} = \begin{bmatrix} \sin(0.1t) \\ \sin(0.1t) \\ \sin(0.1t) \end{bmatrix} \text{ N},$$

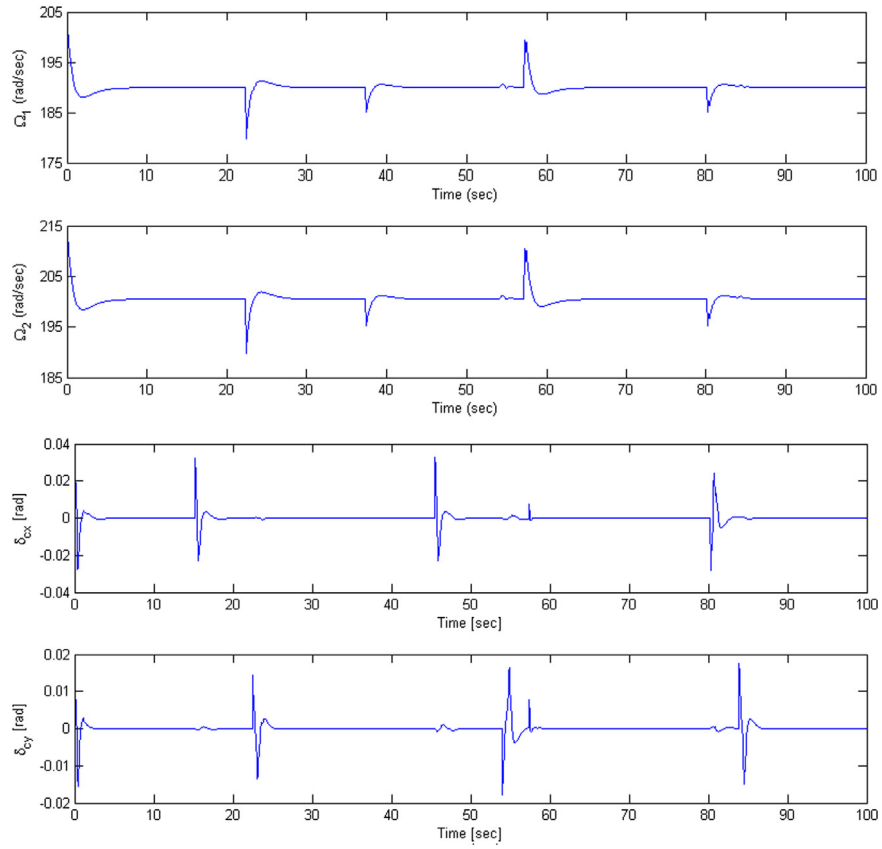


Fig. 7. Case a: control signals Ω_1 , Ω_2 , δ_{cx} and δ_{cy} .

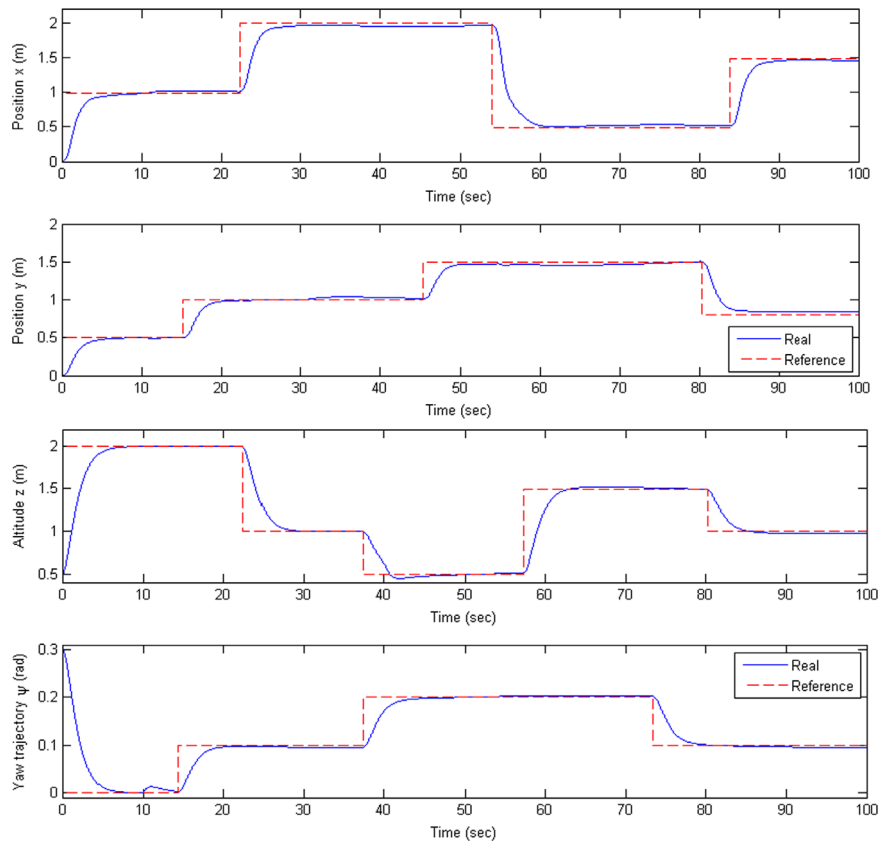


Fig. 8. Case b: position outputs.

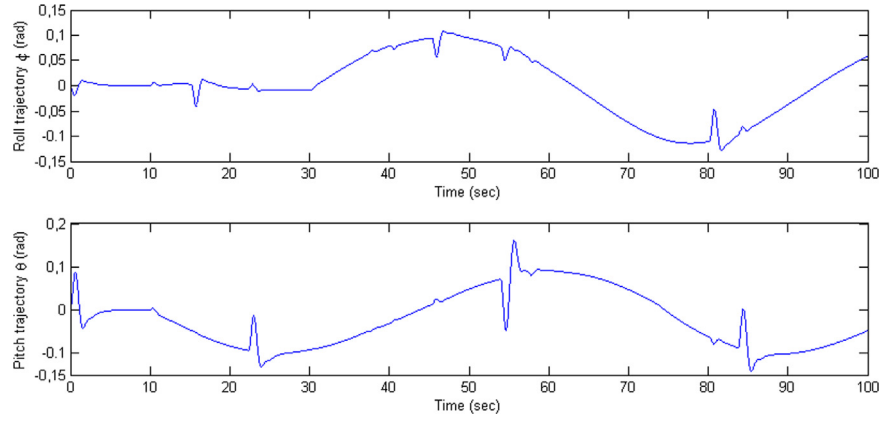


Fig. 9. Case b: roll & pitch trajectories.

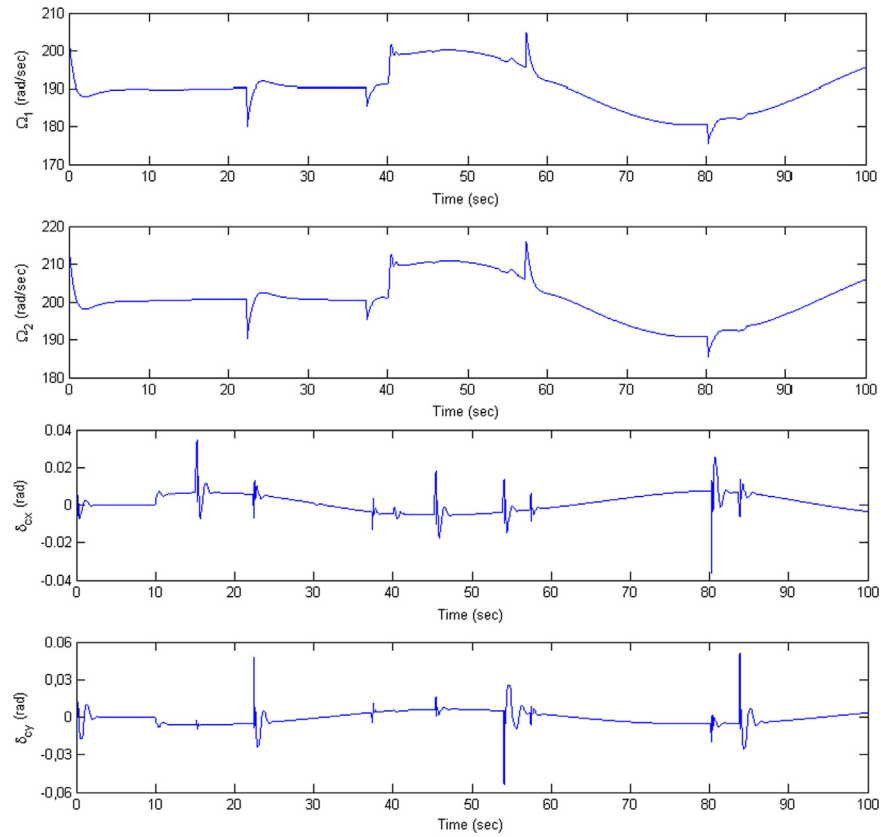


Fig. 10. Case b: control signals Ω_1 , Ω_2 , δ_{cx} and δ_{cy} .

- For the aerodynamic moments occurring at 10 s:

$$\Gamma_{ext} = \begin{bmatrix} 0.3 \sin(0.1t) \\ 0.3 \sin(0.1t) \\ 0.5 \sin(0.1t) \end{bmatrix} \text{ N m}$$

Fig. 8 shows the absolute position of the coaxial-rotor during its flight, when the aerodynamic forces and moments disturbances are introduced, the assigned navigational task are successfully achieved and the reference trajectories are tracked with high accuracy. From these figures, we can see well a good tracking of

the desired trajectories, the controller is able to reject them showing the robustness of the proposed control approach and the stability of the closed loop dynamics is guaranteed. We can notice in Fig. 9 an optimization of tilt angles (Pitch and Roll angles) and consequently the use of minimal energy. Finally, the actuator force commands, displayed in Fig. 10, are continuous as desired and could easily be applied to a real-life model.

4.3. Case c: flight with sensing noise

The same numerical simulation as before was processed by considering the aerodynamic forces and moments disturbances, but considering now a white noise on all the sensors signals. Thus,

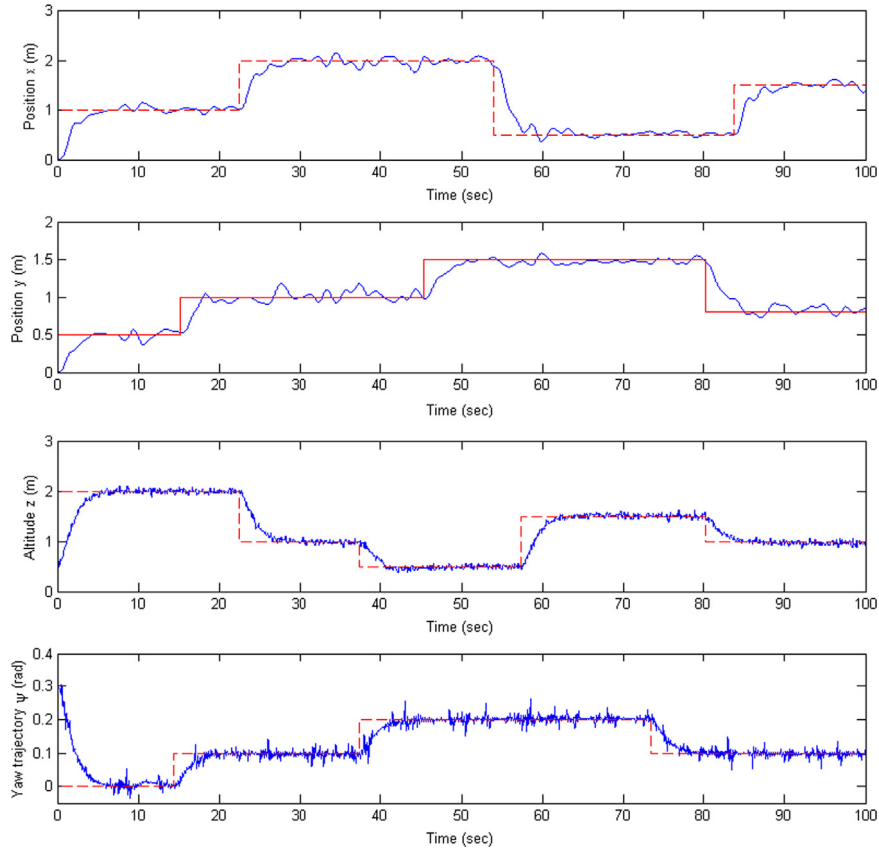


Fig. 11. Case c: position outputs.

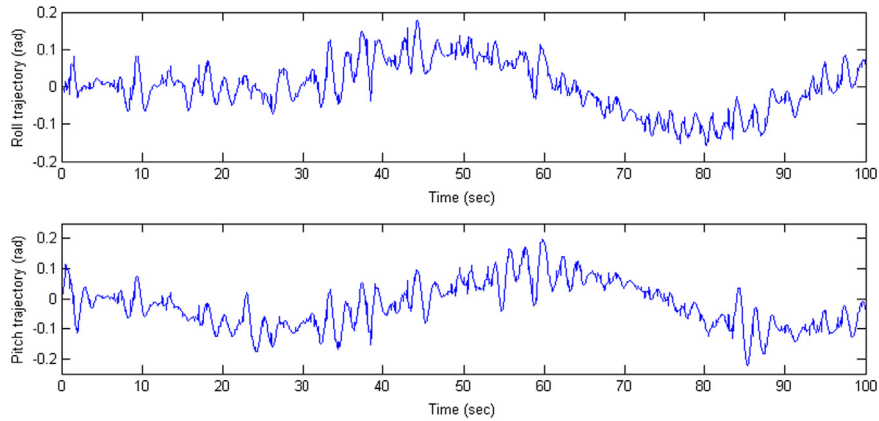


Fig. 12. Case c: roll & pitch trajectories.

realistic uncertain signals are used for control purpose. Despite the sensing noise and the control inputs noise, Figs. 11 and 12 are similar to Figs. 8 and 9, presented in the previous section. From these figures, the designed controller succeeds in reaching the desired position. Nevertheless, depending on the dynamics of the engines, the erratic property of the control signals shown in Fig. 13, which is not a problem here, may be of paramount importance while moving from simulations towards the complex real world scenario of the future experimental validation.

4.4. Case d: comparison with the adaptive backstepping

The efficiency of the proposed control method is illustrated in Fig. 14 via the comparison between the absolute position obtained using the proposed controller and the adaptive backstepping presented in [5] in the presence of uncertainties and external disturbances presented. The simulation results shown in Fig. 14 clearly demonstrate better performances: faster convergence, high-precision tracking and robustness with respect to the adaptive backstepping controller presented in [5].

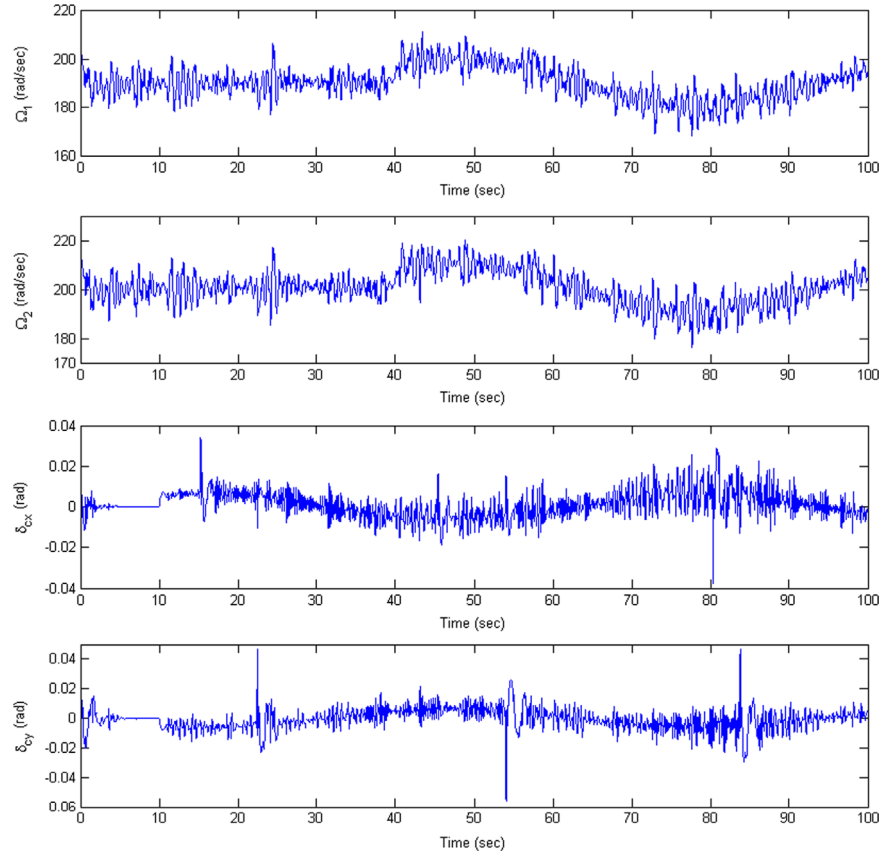


Fig. 13. Case c: control signals Ω_1 , Ω_2 , δ_{cx} and δ_{cy} .

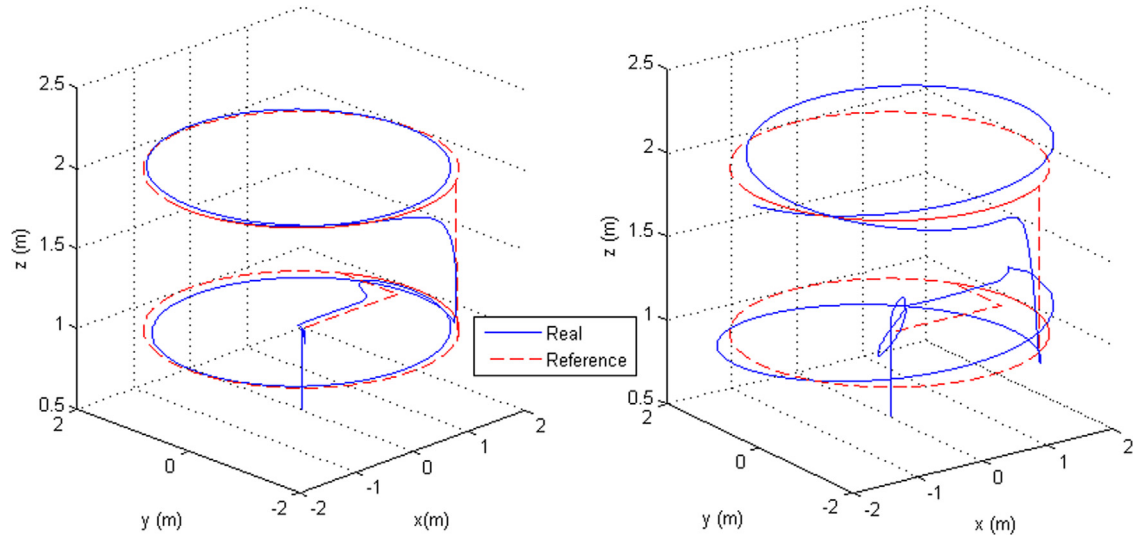


Fig. 14. Case d: comparison between the proposed controller and the adaptive backstepping approach [5].

5. Conclusions

This paper presented a solution to the problem of stabilizing of tracking control of an underactuated coaxial-rotor vehicle along a predefined trajectory in the presence of unknown aerodynamic efforts. A complete dynamics model based on Newton–Euler formalism is designed in the case of quasi-stationary flight. Indeed, we have proposed an original control strategy based on the Linear Extended State Observer – LESO – integrated with the linear controller to solve the path tracking of autonomous coaxial helicopter in

the presence of model uncertainty and external disturbances. Thanks to the dynamical structure model, a nonlinear hierarchical flight control approach is applied. The proposed controller technique combines the advantage of the LESO with the linear controller to force the state variables of the closed-loop system to converge to the desired state and estimates the modeling uncertainty and external disturbance by using LESO. Moreover, the controller has demonstrated good performances and stability of the closed-loop system.

As future work, it would be interesting to design a new mechanism for the small rotorcraft UAV in the case of aggressive

flight. Also, it would be interesting to improve the control system by considering the aerodynamic effects at higher speeds and aggressive maneuvers. Further, the design of a new nonlinear controller to estimate the influence of the unknown external force/torque, unmodelled dynamics and variations of the system dynamics.

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